5.0 Requirements Specification

5.1 - Introduction

The proposed Microforest Connect project is a multi-application project encompassing 1) a data contribution interface, 2) an administrator's portal, and 3) an interactive web page, in service of a network of microforests around the Los Angeles area, most notably in Ascot Hills, where the project gets its namesake. The project as a whole will enable microforest visitors and volunteers to report data on the health and size of the plants, and will aggregate that data into an interactive experience that conveys the microforests' beneficial impact on the environment.

The remaining sections of this document contain a comprehensive breakdown of the Microforest Connect project components and subcomponents (5.2), specific functional requirements that the completed project will fulfill (5.3), specific performance requirements that the completed and optimized project will meet (5.4), and information about the project's environment in both development and usage (5.5).

5.2. - CSCI Component Breakdown

The application is composed of the following CSCs:

5.2.1 - Data Contribution CSC

Application component dedicated to reporting microforest data

- 5.2.1.1 - Log In/Sign Up CSU
    - 5.2.1.1.1 - Application Title (Display title of application)

5.2.1.1.2 - Log In button

5.2.1.1.3 - Sign Up button

5.2.1.1.4 - Log In/Sign Up fields (username, password)

5.2.1.2 - Plant Data Form Tab CSU

5.2.1.2.1 - Data fields (Plant ID, plant health, plant height)

5.2.1.2.2 - QR Code Scanner interface

5.2.1.2.3 - Submit button

5.2.1.3 - Registered Forests Tab CSU

5.2.1.3.1 - Registered forests list

5.2.1.3.2 - Register with new forest Button

5.2.2 - Administrator (Ranger) Portal CSC

Application component dedicated to enabling admins to manage their microforest data.

5.2.2.1 - Admin Sign In CSU

5.2.2.1.1 - Log In/Sign Up fields (email, username, password)

5.2.2.1.2 - Forgot password button

5.2.2.2 - Admin Dashboard CSU

5.2.2.2.1 - Analytics Window (headline statistics and data summary)

5.2.2.2.2 - Users Window (lists users that have contributed)

5.2.2.2.3 - Data Window (lists data that has been gathered)

5.2.2.2.4 - Forest selector (dictates which forests' data is shown)

5.2.2.2.5 - Download as CSV button (allows download

5.2.2.2.6 - Sign-out Button

5.2.3 - Microforest Connect CSC

Visual web interface displaying microforest locations and environmental impact

5.2.2.1 - Map CSU (interactive map of microforests)

5.2.2.1.1 - Microforest Pings (locations of microforests on map)

5.2.2.2 - Microforest Details CSU (details sheet about microforests)

5.2.2.2.1 - Microforest Title

5.2.2.2.2 - Microforest stewards (responsible organization(s))
5.2.2.2.2 - Microforest scale (size)

5.2.4 - Backend Database CSC
The modular data manipulation system represented by the Workflow Editor.
5.2.4.1 - Forests Database CSU
5.2.4.1.1 - Forest Metadata (location, administrators, etc.)
5.2.4.1.2 - Plant Data (individual logs of plant health statuses)

5.2.4.2- Users Database CSU
5.2.4.2.1 - User Metadata (name, email, username)
5.2.4.2.1 - Registered Forests (list of forests the user contributes to)

5.3 - Functional Requirements by CSC

5.3.1 - Data Contribution CSC

5.3.1.1 - Log In/Sign Up CSU

- **5.3.1.1.1** - The application shall display the official application title on the authentication screen.
- **5.3.1.1.2** - The application shall provide input fields for a username and password.
- **5.3.1.1.3** - The application shall provide a Log In button to authenticate existing users.
- **5.3.1.1.4** - The application shall provide a Sign Up button to allow new users to create an account.

5.3.1.2 - Plant Data Form Tab CSU

- **5.3.1.2.1** - The application shall provide data entry fields for Plant ID, plant health status, and plant height.
- **5.3.1.2.2** - The application shall interface with the device camera to provide a QR Code Scanner for automated Plant ID entry.

- **5.3.1.2.3** - The application shall provide a Submit button to upload the completed plant data form to the database.

5.3.1.3 - Registered Forests Tab CSU

- **5.3.1.3.1** - The application shall display a list of all microforests the user is currently registered to contribute to.
- **5.3.1.3.2** - The application shall provide a "Register with New Forest" button to allow users to join additional microforest projects.

5.3.2 - Administrator (Ranger) Portal CSC

5.3.2.1 - Admin Sign In CSU

- **5.3.2.1.1** - The portal shall provide fields for administrative email, username, and password.
- **5.3.2.1.2** - The portal shall provide a "Forgot Password" functionality to trigger account recovery.

5.3.2.2 - Admin Dashboard CSU

- **5.3.2.2.1** - The dashboard shall feature an Analytics Window displaying headline statistics and summaries of collected data.
- **5.3.2.2.2** - The dashboard shall feature a Users Window that lists all contributors associated with the selected forest.
- **5.3.2.2.3** - The dashboard shall feature a Data Window displaying a granular list of all gathered plant logs.
- **5.3.2.2.4** - The dashboard shall provide a Forest Selector to filter the displayed data by specific microforest locations.
- **5.3.2.2.5** - The dashboard shall provide a "Download as CSV" button to export the current data view into a local file.
- **5.3.2.2.6** - The dashboard shall provide a Sign-out button to terminate the administrative session.

5.3.3 - Microforest Connect CSC

5.3.3.1 - Map CSU

- **5.3.3.1.1** - The interface shall provide an interactive world map for navigation.
- **5.3.3.1.2** - The interface shall display "Microforest Pings" representing the geographic locations of registered microforests.

5.3.3.2 - Microforest Details CSU

- **5.3.3.2.1** - The interface shall display the Microforest Title upon selection of a map ping.
- **5.3.3.2.2** - The interface shall display the stewards and organizations responsible for the selected forest.
- **5.3.3.2.3** - The interface shall display the physical scale (size) of the selected microforest.

5.3.4 - Backend Database CSC

5.3.4.1 - Forests Database CSU

- **5.3.4.1.1** - The database shall store forest metadata, including geographic coordinates and assigned administrators.
- **5.3.4.1.2** - The database shall maintain a historical log of plant health statuses and growth metrics indexed by Plant ID.

5.3.4.2 - Users Database CSU

- **5.3.4.2.1** - The database shall store user profile metadata, including name, email, and unique username.
- **5.3.4.2.2** - The database shall maintain a relational link between users and the list of forests they are registered to contribute to.

5.4 - Performance Requirements by CSC

5.4.1 - Time Efficiency

- **5.4.1.1** - The **Data Contribution CSC** shall submit plant health logs to the database in under 2 seconds under standard network conditions.
- **5.4.1.2** - The **Admin Dashboard CSU** shall populate analytics and data summaries within 3 seconds of selecting a specific forest.
- **5.4.1.3** - The **Microforest Connect CSC** map interface shall render all microforest pings within 1.5 seconds of the page loading.

5.4.2 - Resource and Memory Efficiency

- **5.4.2.1** - The **QR Code Scanner interface** shall utilize a non-invasive amount of system memory to prevent application crashes on mobile devices.
- **5.4.2.2** - The **Administrator Portal CSC** shall optimize CSV exports to ensure file sizes do not exceed 50MB for standard monthly data logs.
- **5.4.2.3** - The **Backend Database CSC** shall implement data indexing to ensure query times do not increase exponentially as the number of plant logs grows.

5.4.3 - Data Integrity and Replicability

- **5.4.3.1** - The **Forests Database CSU** shall ensure that plant health data remains consistent and replicable across both the Admin Dashboard and the Microforest Connect interface.
- **5.4.3.2** - The **Data Contribution CSC** shall prevent the submission of duplicate Plant IDs within the same submission session to maintain data accuracy.

5.4.4 - Security and Data Privacy

- **5.4.4.1** - The **Users Database CSU** shall encrypt all user passwords and personal metadata to prevent unauthorized access.
- **5.4.4.2** - The **Admin Sign In CSU** shall automatically terminate a session after 30 minutes of inactivity to protect sensitive forest data.

- **5.4.4.3** - The system shall require a secondary confirmation prompt before an administrator deletes a microforest profile or a user account.

5.4.5 - User Interface Intuition and Accessibility

- **5.4.5.1** - The **Plant Data Form Tab** shall be designed for high visibility to ensure ease of use for volunteers working in outdoor, high-glare environments.
- **5.4.5.2** - The **Microforest Connect CSC** shall provide intuitive navigation (drag-and-click) with a learning curve low enough for public users to find forest details without a tutorial.
- **5.4.5.3** - The **Admin Dashboard** shall provide clear error messaging if a CSV download fails or if network connectivity is lost.

5.5 - Project Environment Requirements

5.5.1 - Hardware Requirements

The microforest application requires mobile devices for field data contribution and a web interface for administrative management.

Category	**Requirements**
Mobile Processor	8-core 2.0 GHz or faster (for QR scanning/Map rendering)
Mobile Camera	8MP or higher with Auto-focus (Required for CSU 5.2.1.2.2)
Desktop RAM	8 GB minimum (16 GB recommended for Admin Analytics)

Storage	500 MB of available space (Application & local data caching)
Connectivity	4G LTE / 5G or Wi-Fi (Required for real-time database sync)
Display	High-brightness mobile screen (Recommended for outdoor use)

5.5.2 - Software Requirements

The software environment supports cross-platform for the Data Contribution and Admin components.

Category	**Requirements**
Client OS (Mobile)	Android 10+ or iOS 15+
Client OS (Admin)	Windows 10/11, macOS, or Linux (via modern Web Browser)
Web Browser	Chrome, Firefox, or Safari (must support WebGL for mapping)
Database System	Firebase
Design Tools	Figma for UI/UX

5.5.3 - Development Stack & Infrastructure

To develop Ascot, I'll be using the following apps and technologies:

- **Frontend:** React Native (mobile) or React (web) using Tailwind CSS
- **Mapping:** Google Maps API or Mapbox (for Microforest Connect)
- **Backend:** Firebase and Node.js
- **Version Control:** Git and GitHub
- **Deployment:** Combination of Firebase hosting and Expo project hosting